

Natural Language Image Retrieval Using CLIP

Technical Project Proposal

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Date: April 17, 2026

1. Summary

The proposal outlines the design and implementation of a natural language image retrieval system that moves beyond keyword matching by leveraging CLIP (Contrastive Language-Image Pretraining) to map both images and text into a shared embedding space, enabling meaning-based retrieval without requiring manual image annotation.

The Minimum Viable Product (MVP) will demonstrate end-to-end semantic retrieval on a curated 200-image set drawn from the Flickr30k dataset. Performance will be evaluated against a predefined golden query set using Precision@5, Recall@10, and Mean Reciprocal Rank.

The project further identifies a prioritized set of stretch goals including an interactive Streamlit web interface, a side-by-side model comparison mode, fine-tuning domain-specific data, and multilingual query support.

2. Problem Statement

Traditional keyword-based image search is constrained by label quality - a photo of a rainy street at night cannot be retrieved unless someone has explicitly tagged it as such. This project replaces keyword matching with semantic retrieval using CLIP, enabling users to query image collections using natural language without requiring manual annotation.

2.1 Target Use Cases

- Journalists searching news photo archives using descriptive queries (e.g., "protest in a public square at dusk")
- Researchers retrieving scene-specific images from datasets without manual annotation
- Content creators locating concept-based stock imagery by description
- Visually impaired users searching image libraries using natural language descriptions instead of visual browsing

3. Technical Pipeline

The system integrates three components - Computer Vision, Natural Language Processing, and vector similarity search - into a unified retrieval pipeline.

3.1 Component Overview

Computer Vision - CLIP encodes images into 512-dimensional embedding vectors. CLIP was selected because it was explicitly trained on image-text pairs, making its visual embeddings directly comparable to text embeddings in the same space without a separate cross-modal alignment step.

Natural Language Processing - The CLIP text encoder transforms free-form user queries into vectors in the same 512-dimensional embedding space, enabling direct semantic comparison with image embeddings.

Vector Search - Cosine similarity is computed between the query embedding and all stored image embeddings, returning the top K for most semantically relevant images. For a dataset of 200 images, exhaustive cosine similarity is computationally sufficient without requiring approximate indexing.

3.2 End-to-End Pipeline

Step	Stage	Description
1	Load Dataset	Flickr30k images loaded from local filesystem
2	Check Embeddings	Verify stored embeddings match current model config; re-embed automatically if model has changed
3	Encode Images	CLIP converts each image to a 512-dim embedding
4	Store Embeddings	Image embeddings saved locally as a model specific .npz file for reuse across sessions
5	Encode Query	User's natural language query passed through CLIP text encoder
6	Compute Similarity	Cosine similarity between query embedding and all image embeddings
7	Retrieve Results	Top K most similar images returned and displayed

4. Datasets

The MVP uses a curated set of 200 images drawn from the Flickr30k dataset - a publicly available collection of 31,000 real-world photographs covering diverse human activities and social scenes, well suited for natural language query evaluation. The dataset is licensed for non-commercial research and educational use.

The subset is intentionally selected to cover diverse scene types including human emotional situations, social gatherings, sports, performance, and animals - ensuring sufficient variety for meaningful semantic retrieval and unambiguous golden query evaluation.

MS-COCO 2017, a large-scale dataset of over 118,000 images across 80 object categories, is identified as a secondary dataset for later project phases where broader object category coverage may be needed.

5. Evaluation

5.1 Golden Query Set

Five predefined natural language queries serve as the evaluation benchmark. Each query has a known set of relevant images within the 200-image corpus, established by manual visual inspection of the dataset. Queries are designed around scene types and activities that are well-represented in the dataset ensuring that expected results are unambiguous, and evaluation is meaningful.

Q1 "a person playing tennis on a court": 22928793.jpg, 22930048.jpg, 23008340.jpg, 23018702.jpg

Q2 "a musician performing with a guitar": 21954377.jpg, 12243003.jpg, 16151663.jpg

Q3 "a group of friends laughing together indoors": 1440465.jpg, 2148982.jpg, 3996401.jpg, 48909501.jpg, 6901333.jpg

Q4 "a child playing outdoors": 10404007.jpg, 15875060.jpg, 16495609.jpg, 6276165800.jpg, 3662865.jpg, 29850055.jpg

Q5 "a group of performers on a stage": 20977655.jpg, 24183660.jpg, 2760167.jpg, 6563243819.jpg, 6787698771.jpg

5.2 Metrics and Target Thresholds

Metric	Description	Target Threshold
Recall@10	Fraction of relevant images appearing in top 10 results	> 0.70
Precision@5	Fraction of top 5 results that are relevant	> 0.60
MRR	Average reciprocal rank of the first relevant result across all queries	> 0.60

7. Stretch Goals

The following enhancements are listed in priority order, contingent on successful MVP completion and available time.

- Priority 1: Interactive web interface built with Streamlit - a lightweight Python framework for building browser-based interfaces - enabling real-time query input and visual display of retrieved images with similarity scores
- Priority 2: Side-by-side model comparison mode, running the same query simultaneously on ViT-B-32 and ViT-L-14 and displaying results in parallel to evaluate the impact of model size on retrieval quality
- Priority 3: Fine-tuning CLIP on domain-specific data to improve retrieval accuracy for targeted use cases
- Priority 4: Multilingual query support to allow searches in languages other than English